GG-H-216C
October 22, 1976
SUPERSEDING
Fed. Spec. GG-H-216B
September 29, 1960

FEDERAL SPECIFICATION

HEMACYTOMETER SET

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a hemacytometer set, complete, in carrying case, as well as the individual components.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566--Boxes, Folding, Paperboard.
PPP-B-585--Boxes, Wood, Wirebound.
PPP-B-601--Boxes, Wood, Cleated-Plywood.
PPP-B-621--Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636--Boxes, Shipping, Fiberboard.
PPP-B-676--Boxes, Setup.

Federal Standards:

Fed. Std. No. 123--Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547--Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-T-36966--Tubing, Rubber.

Military Standards:

MIL-STD-105--Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129--Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Department of Commerce, National Bureau of Standards Publication:

Scientific paper No. 507--A New Interference Apparatus for Testing Hemacytometers.
Scientific paper No. 579--Testing of Cover Glasses for Hemacytometers.

Circular 602--Testing of Glass Volumetric Apparatus.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. NOTE: Scientific paper No. 507 is out of print and is no longer available from the Superintendent of Documents.)

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American Society for Testing and Materials (ASTM) Standards:

C162--Standard Definitions of Terms Relating to Glass and Glass Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Hemacytometer set. Unless otherwise specified, each hemacytometer set shall consist of the following items, in the quantities specified (see 6.1):

1 Counting chamber, double.
2 Cover glasses, light (0.4 millimeter (mm.) thickness).
1 Pipet, diluting, for red blood corpuscles (with rubber tubing and mouthpiece).
1 Pipet, diluting, for white blood corpuscles (with rubber tubing and mouthpiece).
1 Sheet of instructions.
1 Carrying case.

3.2 Material.

3.2.1 Counting chamber. The counting chamber shall be of a low-expansion borosilicate or equal glass, free from optical imperfections in the ruled area.

3.2.2 Cover glasses. Material shall be colorless, transparent glass (see 6.1).

3.2.3 Pipets.

3.2.3.1 Glass. Diluting pipets shall be made of high quality, clear, transparent glass.

3.2.3.2 Rubber tubing. The rubber tubing shall conform to the requirements of MIL-T-36966 for light-wall, pure gum tubing.

3.2.4 Carrying case.

3.2.4.1 Plastic. Plastic carrying case shall be made of semi-flexible polyethylene, impervious to xylol, and similar solvents ordinarily used in hematology laboratories.

3.2.4.2 Wood. Wood carrying case shall be made of well-seasoned, kiln-dried, sound hardwood, which will not warp or split with age or use.

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3.2.4.3 Metal. Metal carrying case shall be made of a good commercial quality corrosion-resistant metal.

3.3 Counting chamber. The counting chamber shall be constructed of one-piece glass with an H-shaped moat forming two counting areas with improved double Neubauer rulings and shall measure 75.5 ± 1.5 mm. in length, 33 ± 0.8 mm. in width, and not less than 5 mm. in thickness. Equidistant from each end of the slide, two transverse elevated ribs (moat walls), 4.5 ± 0.8 mm. in width at the base, shall extend the width of the slide to serve as supports for the cover glass. An elevation, forming the base of the cell chamber, shall be situated centrally between the coverglass supports. The ends of the elevation shall be on a line with the ends of the cover-glass supports. The elevation shall be 5.9 ± 1.5 mm. wide and shall be divided at its center by a transverse trench 4.2 ± 1.4 mm. wide. The counting chamber shall meet the requirements in the following subparagraphs when tested as specified in 4.4.1.1.

3.3.1 Cell chamber. The open-type cell chamber, resulting from the above-described elevations, shall be not less than 0.5 mm. and not more than 1.5 mm. above the surface of the slide.

3.3.1.1 Depth. The depth of each of the two cell chambers, formed by placing an optically plane coverglass on the supports, shall be 0.1 mm. ± 0.002 mm. when tested as specified in 4.4.1.1. The distance between the outer edges of the coverglass supports shall be greater than 20 mm.

3.3.1.2 Ruling. An area 3.0 mm. square shall be ruled into nine equal 1.0 mm. ± 0.01 mm. squares on both cell chambers, in accordance with the accepted Neubauer ruling. The ruled area shall be located on the interior half of the elevation. The central square millimeter shall be divided into 400 squares of 1/400 square millimeter each; every fourth line shall be a double or triple line. Of the eight surrounding 1.0 mm. squares, four corner squares shall be subdivided into sixteen equal squares and the other four squares into twenty equal squares. The four middle surrounding 1.0 mm. squares may have additional lines through one dimension as a result of extending the rulings of the central square millimeter to the edges of the area. The ruling shall be type 1 or 2 described in 3.3.1.2.1 and 3.3.1.2.2, unless one particular type is specified in the invitation for bids or request for proposals (see 4.4.1.1 and 6.1).

3.3.1.2.1 Type 1 ruling. The ruling shall consist of lines ruled through a semi-transparent, bright metal-coated surface which has been deposited on the glass counting area. The metal coating shall be bonded to the glass.

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3.3.1.2.2 Type 2 ruling. The ruling shall consist of sharply defined, permanent lines cut or etched directly into the glass surface of the counting area.

3.3.2 Planeness. The surface supporting the cover glass shall be plane within one wavelength of sodium light over the portion in contact with the cover glass when tested as specified in 4.4.1.1. The surfaces shall have a matte finish. If specified in the invitation for bids or request for proposals, the surfaces shall be polished sufficiently to secure Newton color fringes between the cover glass and the supports. The polished ruled surface shall be planed within two wavelengths of sodium light (see 4.4.1.1 and 6.1).

3.3.3 Concavity. The surface underneath the chamber shall have a cavity of sufficient size and depth to protect the undersurface immediately beneath ruled areas from becoming scratched.

3.3.4 Marking. Each counting chamber shall be permanently marked, indicating the type of ruling, depth of chamber, and any additional information routine with the manufacturer.

3.4 Cover glass.

3.4.1 Shape and dimensions. Cover glass shall be rectangular or octagonal in shape, 20 mm. $\pm$ 1.0 mm. by 26 mm. $\pm$ 1.0 mm. with a thickness of 0.4 mm. $\pm$ 0.06 mm. Thickness shall be tested as specified in 4.4.2.1.

3.4.2 Solubility. No pink color shall be visible through 24 superimposed cover glasses when tested as specified in 4.4.2.2.

3.4.2 Planeness. The surface of the cover glass shall not deviate on both sides from a true plane by more than 0.002 mm. when tested as specified in 4.4.2.3.

3.5 Pipets, diluting. Diluting pipets shall consist of a small, uniform-bore, glass tube which shall have a bulb of proper size near the proximal end. As an alternate, the pipet may be constructed from two halves fused together to form a bulb of a design that is other than the usual round or oval shape. The bulb may be slightly frosted but still transparent. The capacity of the bulb shall be 0.8 milliliter (ml.) to 1.2 ml. for the red pipet and 0.2 ml. to 0.4 ml. for the white pipet. The bulb shall contain a small, free, non-spherical, glass or plastic bead, sufficiently large to prevent its being impacted in the constricted portion of the bulb. The cross section of the pipet at any point shall be circular. The distal end of the white pipet (see 3.5.6.1) shall be drawn down, and the end shall be carefully ground and polished to a short taper point. The distal end of the red pipet (see 3.5.6.2)

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shall be ground to a taper, and the taper shall be carefully polished.

As an alternate, the distal end of the red pipet may be drawn down the end and carefully ground and polished. The inside diameter at the end of both pipets shall be between 0.30 mm. and 0.50 mm. The stem below the bulb shall have two or ten divisions. If there are two divisions, they shall be at the 0.5 and 1.0 volume ratios of the total capacity; if there are ten divisions, there shall be ten equal spaces with the fifth and top lines (below bulb) numbered 0.5 and 1.0 respectively. Total capacity for the white pipet shall be 11 volumes and for the red pipet 101 volumes. The graduation indicating the total shall be etched or fused on above the bulb (see 6.1). The length of the pipets shall be not less than 104 mm. nor more than 121 mm. The graduation marking line for "1" shall be located not less than 2 mm. nor more than 5 mm. from the end of the bulb or straight-bore part of the capillary. The top line (marked 101 on the red pipet and 11 on the white pipet) shall be located not less than 2 mm. nor more than 4 mm. above the mixing chamber or bulb. The diameter of the bore shall be uniform from the bulb to the top line.

3.5.1 Rubber tubing. Each pipet shall be fitted with a section of rubber tubing approximately 10 inches long, having an inside diameter of 1/8 inch and a wall thickness of 1/32 inch.

3.5.2 Mouthpiece. The mouthpiece shall be fabricated of white or natural color, chemically-resistant plastic or bone, and shall be double-flanged, round tube type, or wedge-shaped design at the mouth (proximal) end. The outside diameter of the distal end of the mouthpiece shall be such that it will fit snugly into the 1/8 inch inside diameter latex rubber tubing described in 3.5.1. The mouthpiece shall be suitable for use in applying suction or pressure to the blood-diluting pipets without slippage on any size lip.

3.5.3 Graduation marking. All graduations shall be at right angles to the long axis and parallel with each other. Graduations shall be marked with proper numerical value and shall be not less than two thirds of the circumference.

3.5.3.1 Diluting pipet for white blood cells. The graduation markings of the one-half (1-20 dilution) volume, the one (1-10 dilution) volume, and the total capacity (11 volumes) shall be clearly designated by Arabic numerals.

3.5.3.2 Diluting pipet for red blood cells. The graduation markings of the one-half(1-200 dilution) volume, the one (1-100 dilution) volume, and the total capacity (101 volumes) shall be clearly designated by Arabic numerals.

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3.5.4 Identification marking. Each pipet shall have an identifying serial number, and the name or registered trademark of the manufacturer or distributor.

3.5.5 Pigmentation. All markings shall be fused on with a permanent dark color. When tested in accordance with 4.4.3.1, the pigmentation shall not discolor the test solution. As judged by the eye, the appearance of all color markings shall be the same after testing as before.

3.5.6 Identifying color.

3.5.6.1 Diluting pipet for white blood cells. Each pipet shall have a milk-glass background or milk-glass (or white plastic) bead, or both to identify it readily as a white pipet.

3.5.6.2 Diluting pipet for red blood cells. Each pipet shall have a ruby-glass background, or ruby-glass (or red plastic) bead, or both to identify it readily as a red pipet.

3.5.7 Accuracy. When tested as specified in 4.4.3.2, the ratio of the sample drawn to the bulb volume shall be accurate within a tolerance of $\pm$ 1 percent.

3.5.8 Thermal resistance.

3.5.8.1 Boiling. The pipets shall show no evidence of corrosion, scumming, chipping or cracking, when tested in accordance with 4.4.3.3.1.

3.5.8.2 Shock. The pipets shall not crack when tested in accordance with 4.4.3.3.2.

3.6 Carrying case. A carrying case, fabricated of plastic, wood or metal as hereinafter specified shall be provided (see 6.1).

3.6.1 Plastic. When plastic is specified, the carrying case shall be made of material specified in 3.2.4.1. The case shall be pocket-size of a pouch-type design with separate compartments for holding the counting chamber, the pipets and a separate sheath for holding the cover glasses. The outside dimensions of the case shall be 5-1/4 $\pm$ 1/8 inches long by 3-3/8 $\pm$ 1/8 inches wide by 7/8 $\pm$ 1/16 inch deep. The case shall be sufficiently rigid to protect the contents from damage and shall have a suitable closure.

3.6.2 Wood or metal. When wood or metal is specified, the carrying case shall be made of material specified in 3.2.4.2 or 3.2.4.3. The case shall be sturdily constructed of wood, with maximum outside dimensions of 7-1/8 inches

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in length, 3-7/8 inches in width and 1-3/16 inches in depth, or metal with maximum outside dimensions of 6-1/2 inches by 3-1/2 inches and 1-1/4 inches in depth. The case shall consist of a body and lid, each of approximately equal thickness, securely hinged in the back and fitted with a suitable bar or catch lock; or if a metal case, it shall have a snap lid. The case shall have rounded corners and beveled edges.

The entire outside surface shall be covered with durable imitation leather, which shall be carried over the edges onto the inner surface. The interior of the case shall be provided with suitable padded recesses or compartments for holding each item in such a manner as to insure retention of the item in its proper compartment or recess when the case is carried in an upright position. The entire inside surface of the case shall be completely lined with plush; or, the inner top may be lined with plush and the inner bottom section of the case may be covered with similar material as used on outside surface, except that compartment at center for counting chamber may be plush-lined.

3.6.3 Identification marking. The cover of each carrying case shall be stamped with the name of the item and the name or registered trademark of the manufacturer.

3.7 Instructions. A complete and detailed set of instructions shall be furnished with each hemacytometer set.

3.8 Workmanship. The workmanship shall be first class throughout. The complete hemayctometer set and individual parts shall be free from defects which detract from the appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the glass and rubber, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificates shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results and the specification requirements.

4.2 Sampling.

4.2.1 Examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. Sampling for examination

	Unit of product	Inspection	AQL (percent defective)
For visual examination			
Major defects	Set	S-4	2.5
	Counting chamber (exclusive of ruled area)	II	2.5
	Ruled area of counting chamber	II	0.65
	Cover glass	II	2.5
	Pipet (red corpuscles)	II	2.5
	Pipet (white corpuscles)	II	2.5
	Case	II	2.5
	Tubing and mouthpiece	II	2.5
Minor defects	Counting chamber (exclusive of ruled area)	II	4.0
	Ruled area of counting chamber	II	1.0
	Cover glass	II	4.0

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TABLE I. Sampling for examination (cont'd.)

	Unit of product	Inspection	AQL (percent defective)
For visual examination			
Minor defects	Pipet (red corpuscles)	II	4.0
	Pipet (white corpuscles)	II	4.0
For dimensional examination	Set	S-3	2.5
	Counting chamber	S-2	2.5
	Pipets	S-2	2.5
	Cover glass (except thickness)	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for tests

Component (Unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
Counting chamber	Depth	3.3.1.1	4.4.1.1	S-1 min. of 2	*
	Ruling	3.3.1.2	4.4.1.1	S-1 min. of 2	*
	Planeness	3.3.2	4.4.1.1	S-1 min. of 2	*
Cover glasses	Thickness	3.4.1	4.4.2.1	S-2	1.5
	Solubility	3.4.2	4.4.2.2	24 cover glasses for each lot shall be used	*
	Planeness	3.4.3	4.4.2.3	S-1 min. of 5	*
Pipet	Permanency of pigment	3.5.5	4.4.3.1	S-2 min. of 10	*
	Accuracy of dilution	3.5.7	4.4.3.2	S-2 min. of 7	*
	Boiling test	3.5.8.1	4.4.3.3.1	S-2 min. of 10	*
	Shock test	3.5.8.2	4.4.3.3.2	S-2 min. of 10	*

* Acceptance number shall be zero for all sample sizes.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

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TABLE III. Classification of defects

Categories	Defects
Major	
101	Component part missing.
102	Set not free of rattle when the case is carried in an upright position with all components inside the case.
103	Case cannot be securely closed with all components inside.
104	Glass components not free of sharp edges, chips, checks and splits.
105	Glass components not free of blisters, tears, blowover and protrusions.
106	Case not free of sharp edges, dents and cracks.
107	Leatherette, polyethylene and plush lining not free of tears and cuts.
108	Leatherette not secured to case.
109	Plush lining not secured to case.
110	Mouthpiece and tubing not free of cracks, rough areas and sharp edges.
111	Pipet stem not parallel to axis of pipet.
112	Pipet bore not free of occlusions, dirt or foreign matter.
113	Pipet identification marking missing, illegible and incomplete.
114	Cover glass not clear (clouded).
115	Cover glass not free of grease, oil, dirt and stain.
Minor	
201	Glass components not free of seeds, stones and bloom.
202	Glass components not free of sleek (ruled area of Chamber, Counting, Blood Cells, Hemacytometer only).
203	Glass components not free of striae (ruled area of Chamber, Counting, Blood Cells, Hemacytometer only).
204	Cover glass not free of scratches and irregular edges.

* Examination shall not be restricted to the classified possible defects listed above. Glass defects are defined in ASTM C162.

4.3.2 Dimensional examination. The set shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

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4.3.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE IV. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Packaging and packing materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3.3.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Counting chamber.

4.4.1.1 Accuracy. Accuracy shall be determined in accordance with methods described in the National Bureau of Standards scientific paper no. 507.

4.4.2 Cover glasses.

4.4.2.1 Thickness. The thickness shall be measured with a standard cover glass gauge or micrometer thickness gauge.

4.4.2.2 Solubility. Twenty four cover glasses shall be cleaned by immersion for one minute in distilled water, removed and air-dried for one minute in a vertical position. This cleaning operation shall be performed three times, using a new quantity of distilled water for each immersion. Half fill an alkali-resistant 600 ml. Erlenmeyer flask with distilled water and boil for ten minutes. Allow

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water to cool for three minutes, and then decant until the volume in the flask is approximately 100 ml. Keeping the flask covered, let stand until the water has cooled to approximately 70°Centigrade (°C.). Add 0.2 ml. of USP standard phenolphthalein solution and continue to cool to 60°C. Add the cover glasses one at a time. The cover glasses shall be superimposed by tilting and gently swirling the flask. Let solution cool toward room temperature without agitation for one hour. Absorption of carbon dioxide from the atmosphere must be avoided during the test.

4.4.2.3 Planeness. The planeness test shall be conducted as specified in NBS scientific paper no. 507 or 579.

4.4.3 Pipets.

4.4.3.1 Permanency of pigment. The pipets shall be immersed in a Chromic Acid Cleaning Mixture * at room temperature. Let stand for 15 minutes. Remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth or cleaning tissue. Repeat once. The appearance of lines and number is to be the same as before the test, as judged by the eye.

* Chromic Acid Cleansing Mixture Formula:

Sodium Dichromate	200 grams
Water	1,000 ml.
Sulfuric	1,500 ml.

4.4.3.2 Accuracy of dilution. Accuracy of dilution shall be determined in accordance with methods prescribed by the NBS circular 602.

4.4.3.3 Thermal resistance.

4.4.3.3.1 Boiling test. The pipets shall be completely immersed in distilled water in a closed vessel having a small steam vent and boiled continuously for six hours, boiling distilled water being added when necessary to compensate for evaporation losses. The pipets shall be suspended in water so that they do not come in contact with the containing vessels.

4.4.3.3.2 Shock test. The pipets shall be immersed in tap water at 18.5° ± 2.5°C. (65° ± 5°Fahrenheit (°F.)), allowed to remain ten minutes, then transferred into boiling water. This operation shall be repeated five times.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.1).

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5.1.1 Levels A and B.

5.1.1.1 Unit packaging.

5.1.1.1.1 Hemacytometer set. Components comprising each set shall be positioned within their respective compartments in the carrying case. The case shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure of the box shall be as specified in the appendix of the applicable box specification.

5.1.1.1.2 Components.

5.1.1.1.2.1 Counting chamber. Each chamber shall be wrapped in tissue paper and packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the box specification.

5.1.1.1.2.2 Cover glass. Each cover glass shall be wrapped in tissue paper and packaged in an envelope of a suitable size. Closure of the envelope shall be adequate to prevent accidental opening.

5.1.1.1.2.3 Mouthpiece. Twelve mouthpieces shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.1.1.3 Pipets (red or white). Unless otherwise specified, each pipet shall be wrapped and cushioned with tissue paper and packaged in a plastic or spiral-wound paperboard tube or in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent spilling of contents.

5.1.1.1.3.1 Pipets. When specified, twelve pipets of one description only shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. The box shall be provided with cushioning material to prevent shifting of or damage to the pipets. The box shall be closed in accordance with the applicable box specification.

5.1.1.1.4 Tubing. Tubing shall be packaged as specified in MIL-T-36966.

5.1.1.2 Intermediate package.

5.1.1.2.1 Hemacytometer set. An intermediate package shall not be required.

5.1.1.2.2 Counting chamber, cover glass, mouthpiece or pipet. Unless otherwise specified, twelve unit packages of one component only shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.1.2.3 Tubing. Tubing shall be packed as specified in MIL-T-36966.

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5.1.2 Level C. Hemacytometer set or individual components, as specified (see 6.1), shall be packaged in a manner that will afford adequate protection against physical damage during shipment from the supply to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Hemacytometer set. Fifty unit packages, containing complete sets, shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, banding and waterproofing shall be as specified in the appendix of the box specification.

5.2.1.2 Components.

5.2.1.2.1 Counting chamber, cover glass, mouthpiece and pipet. Unless otherwise specified, twelve intermediate packages containing counting chambers, one hundred forty four intermediate packages containing cover glasses, or forty eight intermediate packages containing mouthpieces or pipets of one type only shall be packed in an exterior container as specified in 5.2.1.1.

5.2.2 Level B. Hemacytometer sets and components shall be packed as specified in 5.2.1.1 and 5.2.1.2.1 except that the exterior container shall conform to PPP-B-636, class domestic and banding and waterproofing shall not be required.

5.2.3 Level C. The packaged hemacytometer sets or individual components shall be packed in shipping containers in a manner that will afford adequate protection against physical damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carrier's rules and regulations.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or purchase order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies.

5.3.2.1 Interior package. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

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5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information on procurement documents:

a. Title, number and date of this specification.
b. Name of component when required (see 3.1).
c. Type of ruling if only one required (see 3.3.1.2).
d. If polished surfaces for cover glass supports are required (see 3.3.2).
e. If graduation indicating total shall be etched or fused on (see 3.5).
f. Material of carrying case (see 3.6).
g. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc. of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc. which are normally procured to meet Federal Government requirements.

Military Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6630-0271

Civil Agency Coordinating Activities:
VA-DMS
HEW-FDA
GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 35 cents each.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

GG-H-216C - Hemacytometer Set

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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